

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – DEC 2011

PT108 FUNDAMENTALS OF BIOMECHANICS

Date: 05.12.11, Monday

Time: 10.00 a.m to 10.20 a.m

Maximum Marks: 10

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(1 × 10 = 10)

1. Center of mass in anatomic standing position lies approximately
 - a. At S2 level.
 - b. Anterior to S2 level.
 - c. Posterior to S2 level.
 - d. At pubic symphysis.
2. The following is true for greater stability EXCEPT
 - a. Large base of support.
 - b. Center of mass nearer to base of support.
 - c. Center of mass far from base of support.
 - d. Line of gravity within base of support.
3. Which one of following receptor is sensitive to pressure and forceful joint motion in to extremes of motion?
 - a. Ruffini.
 - b. Pacini- form.
 - c. Golgi- Mazzoni.
 - d. Unmyelinated free nerve endings.
4. Following of these are Synchondrosis EXCEPT.....
 - a. Costochondral joint.
 - b. Manubriosternal joint.
 - c. Costovertebral joint.
 - d. Xiphisternal joint.
5. Which of the following muscle is required for quiet expiration?
 - a. Diaphragm.
 - b. Pectoralis major.
 - c. Internal Intercostal.
 - d. None of the above.

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Section	_____
Date	_____
Sr. Supervisor's Sign.	

6. The net force that moves a bony segment away from its adjacent bony segment is known as
- Distraction force.
 - Joint reaction force.
 - Gravity force.
 - All of the above.
7. Strength of rotation produced by force couple is known as
- Torque.
 - Moment of force.
 - Moment of arm.
 - Both a or b.
8.cells are responsible for bone resorption.
- Osteoblast.
 - Osteoclast.
 - Fibroblast.
 - Fibrocytes.
9. Which of the following is helpful in Extension of Neck EXCEPT?
- Splenius Capitis
 - Splenius Cervicis
 - Erector spinae
 - Sternocleidomastoid
10. The lower joint formed by mandible condyle & inferior of disk istype of joint.
- Plane
 - Saddle
 - Ball & socket
 - Hinge

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PT108 FUNDAMENTALS OF BIOMECHANICS

Date: 05.12.11, Monday

Time: 10.20 a.m to 1.00 p.m

Maximum Marks: 60

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2)

(1× 10= 10)

Describe lever system with examples and explain mechanical advantage tradeoffs and limitations of lever system.

OR

Explain the different types of joint structure.

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Intervertebral Disc.
2. Anatomical pulleys.
3. Composition of Forces with example of each.
4. Factors affecting muscle function.
5. Stress strain curve.

SECTION –II

Q.3. Long Question (1 out of 2)

(1× 10= 10)

Explain biomechanics of Cervical spine.

OR

Define Posture. Explain ideal posture analysis from lateral view (in saggital plane).

Q.4. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Active Insufficiency.
2. Motor unit.
3. Mechanism of respiration.
4. Muscles of mastication.
5. Lumbar-pelvic rhythm.